

Residual-driven sensitivity analysis for pressure drop prediction in packed beds of spherical particles

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The accurate prediction of pressure drop across randomly packed beds is critical for the design of chemical reactors, adsorption columns, and thermal energy storage systems, yet existing semiempirical correlations often lack consistent accuracy across diverse geometric and flow regimes. This study presents a residual-driven sensitivity analysis framework that systematically evaluates the influence of key operational and geometric parameters, namely, superficial velocity U_s , normalized bed length L/D , and normalized column diameter D/d_p , on the prediction error of classical pressure drop models. Using an extensive experimental dataset collected in our laboratory, comprising over 3000 measurements across four column diameters (50–200 mm) and six particle sizes (2.5–42.6 mm), we integrate robust outlier detection, iterative data sanitation, and multimethod sensitivity assessment combining Pearson correlation, tree-based feature importance, and partial dependence analysis. Results show that for the raw dataset, errors are collectively influenced by all three parameters, with no single factor dominating. Through successive cleaning iterations that progressively remove data with strong wall effects ($D/d_p < 4$), low length-to-diameter ratios, and extreme geometries, the analysis reveals a clear transition: Superficial velocity emerges as the overwhelmingly dominant driver of residual error, accounting for 74.6% of the normalized Gini importance in the final filtered dataset. This evolution demonstrates that after mitigating geometric confounders, the core limitation of widely used models such as the Ergun equation lies in their inability to accurately capture flow-regime-dependent behavior. The work provides a validated methodology for sensitivity analysis in packed-bed flows and offers practical guidance for model selection and experimental data-quality assurance, highlighting that future model improvements should prioritize velocity-dependent corrections rather than further refinement of geometric terms.

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I. INTRODUCTION

The accurate prediction of pressure drop in packed beds is a fundamental challenge in chemical engineering, with direct implications for the design and optimization of reactors, adsorption columns, filtration systems, and thermal energy storage units [1,2]. Packed beds, consisting of

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randomly arranged particles, offer highly tortuous flow paths that enhance heat and mass transfer, making them essential in applications such as air purification, flue gas recovery, nuclear reactors, and heat exchangers [3–5]. However, the pressure drop across such beds significantly impacts energy consumption, operational costs, and system efficiency, necessitating reliable predictive models.

The pressure drop in packed beds is described by the Darcy-Forchheimer equation, which expresses the pressure loss as the sum of viscous and inertial contributions

$$\frac{\Delta P}{L} = a\mu U_s + b\rho U_s^2, \quad (1)$$

where a and b are empirical coefficients dependent on bed geometry, particle characteristics, and flow conditions and U_s is the superficial velocity.

Over the past century, numerous researchers have sought to refine coefficients a and b to develop robust correlations applicable across a wide range of operating regimes [6,7].

Early work by Blake [8] introduced the modified pressure drop coefficient f_k and Reynolds number Re_H , framing the pressure loss as a combination of viscous and kinetic energy losses. Carman [1] later proposed the widely used Carman-Kozeny correlation

$$f_k = \frac{180}{\text{Re}_H} + 2.87 \text{Re}_H^{-0.1}, \quad (2)$$

which performs well under moderate Reynolds numbers but shows limitations under extreme geometric or flow conditions.

A seminal contribution was made by Ergun [2], who unified previous observations into the Ergun equation

$$\frac{\Delta P}{L} \frac{d_p}{\rho U_s^2} \frac{\varepsilon^3}{1 - \varepsilon} = 150 \frac{1 - \varepsilon}{\text{Re}_H} + 1.75. \quad (3)$$

While extensively used, the Ergun equation tends to overpredict pressure drop at low tube-to-particle diameter ratios $D/d_p < 10$, where wall effects become significant [9,10].

Subsequent studies aimed to address these limitations. Brauer [11] introduced corrections for void fraction and particle arrangement, while Handley [12] considered packing regularity effects. Montillet *et al.* [13] incorporated the diameter ratio D/d_p as a correcting factor for beds with pronounced wall effects. More recently, Dixon [14] proposed a correlation valid across a broad Reynolds number range from 10^{-2} to 5×10^5 and Papkov *et al.* [5] investigated the effect of particle surface roughness, demonstrating increases in pressure drop by 50%–100% compared to smooth particles under certain conditions.

Despite these advances, existing correlations often exhibit inconsistent accuracy when applied to independent datasets, with mean absolute relative errors ranging to 600% in turbulent flow and 800% in laminar flow [15]. While established semiempirical correlations such as the Carman-Kozeny and Ergun equations are widely used, their predictive reliability can vary significantly under different geometric and flow conditions, particularly when the ratio of column-to-particle diameter D/d_p is low or when bed porosity and packing structure deviate from idealized assumptions. Many correlations are constrained by the limited experimental conditions under which they were validated, restricting their generalizability. Furthermore, traditional sensitivity analyses tend to rely on simulated or highly controlled data, often overlooking measurement uncertainty, data outliers, and nonlinear parameter interactions inherent in real-world experiments.

In this work we present a residual-driven sensitivity analysis framework to systematically evaluate the influence of key operational and geometric parameters, namely, the superficial velocity U_s , normalized bed length L/D , and normalized column diameter D/d_p , on the prediction error of pressure drop models. Using an extensive experimental dataset collected in our laboratory, comprising 95 series of measurements across four column diameters and variable particle sizes, we integrate robust outlier detection, data sanitation protocols, and multimethod sensitivity assessment. An “experimental series” refers to a unique fixed geometric configuration of the packed bed, i.e., a

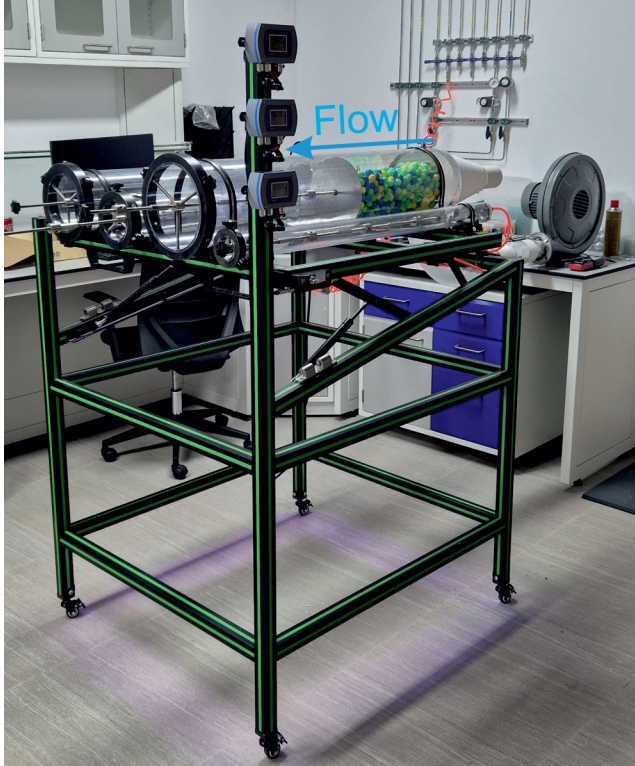


FIG. 1. Schematic overview of the experimental apparatus for pressure drop measurements in randomly packed beds: four transparent polycarbonate columns (inner diameters of 50, 100, 150, and 200 mm and length of 1000 mm) and three differential pressure sensors (200 Pa, 1 kPa, and 10 kPa full scale) to measure the pressure drop across the beds.

specific combination of column diameter D , bed length L , and particle diameter d_p . For each of the 95 series, the pressure drop was measured across a wide range of superficial velocities U_s .

Our approach combines Pearson correlation analysis to quantify linear relationships, tree-based feature importance ranking to assess nonlinear influences, and partial dependence analysis to isolate marginal effects of individual parameters.

Through iterative data filtering and residual mapping, we identify regions of parameter space where classical models such as Carman-Kozeny and Ergun perform poorly, and we quantify the relative importance of each parameter in driving prediction error. The results highlight the dominant role of superficial velocity in nonlinear residual behavior while also revealing how geometric factors gain influence under specific experimental constraints.

This work provides a validated methodology for sensitivity analysis in packed-bed flows and offers practical guidance for model selection and experimental data-quality assurance in similar contexts. The findings contribute to the ongoing effort to improve the accuracy and robustness of pressure drop correlations for industrial applications, where reliable performance across a wide range of operating conditions is essential.

II. METHODOLOGY

A. Experimental setup

The experimental apparatus (Fig. 1) is designed to investigate pressure drop dynamics in randomly packed beds under varying flow conditions. The core of the setup features four transparent

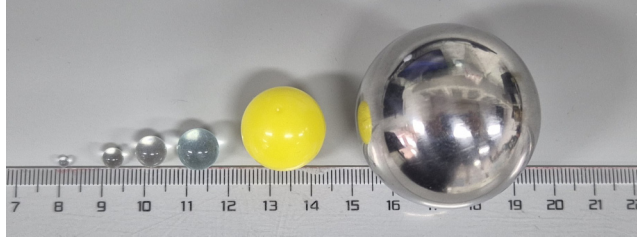


FIG. 2. Spherical particles used in experiments: $d_p = 2.5, 5.0, 7.5, 10, 20$, and 42.6 mm.

polycarbonate pipes, each 1 m long, with inner diameters of 50, 100, 150, and 200 mm (wall thickness of 5 mm). Each pipe accommodates a packed bed of up to 800 mm in length, confined between two identical perforated plates (hole diameter of 2 mm). The fixed inlet plate is located 100 mm from the pipe inlet, forming an inlet plenum for uniform airflow distribution. The movable outlet plate is attached to a threaded rod. Its position is precisely adjusted using five-way nuts and secured with a tripodlike fixation system. This design enables accurate, reproducible compression of the packed bed to any desired length L .

Air is supplied by a 750-W centrifugal blower (model CZ-TD750W), which delivers a maximum flow rate of $1060 \text{ m}^3/\text{h}$ at a pressure head of 2.3 kPa through an 80-mm flexible spiral duct. A secondary low-flow option is provided by a central compressed-air line (maximum $13.2 \text{ m}^3/\text{h}$) connected via an 8-mm polyurethane tube. The system design permits any packed-bed column to be connected to either air source as required.

The pressure drop across the packed bed is measured by three differential pressure sensors [CCY-30, $\pm(0.25\text{--}0.5)\%$ full scale (FS)] with full-scale ranges of 200 Pa, 1 kPa, and 10 kPa. The sensors are connected in parallel to all columns via a manifold of 6-mm polyurethane tubes. A manual valve system selects which column is connected to which sensor. The low-pressure ports of all sensors are vented to the atmosphere, corresponding to the ambient pressure at the column outlets.

Preliminary experiments indicated that the hydraulic resistance of the perforated plates was negligible (less than or equal to 5 Pa) within the considered superficial velocity range of 0.02–4.4 m/s. The resistance was estimated for eight distinct packing configurations, with packing lengths $L = \{100, 200\}$ mm, column-to-particle diameter ratios $D/d_p = \{5, 10\}$, and length-to-diameter ratios $L/D = \{1, 3\}$. For each configuration, the pressure drop was taken as the maximum value from a set of three runs: one with the inlet plate removed, one with the outlet plate removed, and one with both plates installed. The single-plate experiments were conducted with the test pipe oriented vertically to maintain the integrity of the packed bed.

A set of spherical particles was used in the experimental series, with diameters ranging from 2.5 to 42.6 mm (Fig. 2). The 20-mm particles were fabricated from plastic [poly(methyl methacrylate) (PMMA)] for practical and cost reasons, the 42-mm particles were metallic, and smaller glass spheres were commercially available. All particles had a polished surface finish ($R_a < 0.1 \mu\text{m}$). The surface roughness specification was provided by the manufacturers (for glass and PMMA spheres) and confirmed qualitatively using optical microscopy prior to experiments. For the flow regimes studied (Re_p up to 10^5), the pressure drop is dominated by form drag and skin friction, which are functions of geometry and surface roughness, not material density. The sphericity and size uniformity of the particles were characterized by analyzing 20 randomly selected particles from each size fraction. For each particle, 12 diameter measurements were taken at random orientations using an analog precision micrometer with an accuracy of ± 0.01 mm. The mean coefficient of variation in diameter for a single particle was 0.3% for glass beads, 1.1% for PMMA spheres, and 0.5% for metal spheres. The mean coefficient of variation for a given particle sample was 0.3% for glass beads, 0.9% for PMMA spheres, and 1.2% for metal spheres. Given the measured sphericity and comparable smoothness, the material difference is not expected to significantly influence the dimensionless pressure drop correlation.

TABLE I. Design and operating parameters of the experiment.

Parameter	Available range
D (mm)	50–200
d_p (mm)	2.5–42.6
L (mm)	122–805
N_p	11–70 283
ε	0.37–0.62
U_s (m/s)	0.02–4.4
Re	5.5–21 264
L/D	0.63–15.46
D/d_p	1.19–40

The particle sizes (2.5–42.6 mm) and column diameters (50–200 mm) were chosen to achieve a wide range of key dimensionless ratios: D/d_p from approximately 1.2 to 30, covering strong to negligible wall effects. Monodisperse spheres were used to isolate the effect of geometry from polydispersity. Material selection was based on commercial availability of precise spheres; all were smooth to minimize the confounding effect of roughness.

B. Experimental procedure

The investigation involved 95 experimental series defined by the packed-bed column diameter D , length L , and particle diameter d_p . The volumetric flow rate G_v was varied across each series through one of two methods: controlling a blower with a single-phase inverter (AS2-107) at frequencies ranging from 10 to 50 Hz or regulating a manual valve on the compressed air feed line. The measured flow rates were converted to superficial velocities U_s . Derived parameters, namely, the particle count N_p and bed porosity ε , were subsequently calculated. The obtained ranges of design and operating parameters are listed in Table I.

Superficial velocity U_s was calculated from anemometer readings (DLY-1603A, $\pm 1\%$ FS) while using the blower, according to $U_s = U(57/D)^2$, where the vane probe ($D_{\text{vane}} = 57$ mm) was mounted in the outlet plenum of the same diameter. For the compressed air supply, it was calculated from thermal mass flow meter readings (TC5200-200L-A, $\pm 2\%$ FS) as $U_s = 4G_v/\pi D^2$.

Test tubes were randomly packed with moderate manual compaction, resulting in loosely packed beds with porosity of $\varepsilon = 0.37$ – 0.62 . Larger particles were enumerated manually, whereas smaller ones were quantified using a gravimetric method. The mass of a single particle was ascertained by weighing a known number of particles; this value was then used to convert the total sample mass (measured with an HCS3003HD laboratory scale, ± 0.05 g) into a particle count N_p .

C. Sensitivity analysis algorithm

The sensitivity of the model residuals to the input parameters, namely, the superficial velocity U_s , normalized column length L/D , and normalized column diameter D/d_p , was assessed using three complementary techniques. First, Pearson correlation coefficients ρ quantified the linear relationships. Second, normalized feature importance scores ϕ from a tree-based model ranked the parameters' relative influence. Finally, partial dependence analysis was conducted, where $PD(x)$ represents the average change in the residual (in its native units) over the distribution of the parameter x .

We performed z -score normalization [Eq. (14)] on the data features before the main analysis to mitigate artificial bias in the random forest model. This scaling ensures features have a zero mean $\bar{x} = 0$ and unit variance $\sigma_x = 1$. Standardization is necessary because each tree node selects a split based on the condition $x_i \leq t$, where t is the split threshold [16].

The strength and direction of linear relationships between variables were quantified using Pearson correlation coefficients, calculated as [17]

$$\rho_{x_i y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}, \quad (4)$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ is the mean of parameter x_i and $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ is the mean of the target variable.

The feature importance was estimated using a random forest model

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(\mathbf{x}), \quad (5)$$

where h_t are individual decision trees and $T = 100$ is the total number of estimators.

The node splitting criterion, based on Gini impurity, for a node m with N_m samples and K classes was calculated as

$$I_G(m) = \sum_{k=1}^K \hat{p}_{mk}(1 - \hat{p}_{mk}), \quad (6)$$

where $\hat{p}_{mk}$ is the proportion of class k in node m .

For a feature x_i , the Gini importance (feature importance score) was computed as [18]

$$\mathcal{I}(x_i) = \frac{1}{T} \sum_{t=1}^T \sum_{m \in M_t^{(i)}} \frac{N_m}{N} \Delta I_G(m), \quad (7)$$

where $M_t^{(i)}$ is the set of nodes in tree t where x_i is used for splitting, $\Delta I_G(m) = I_G(m) - \frac{N_{m_L}}{N_m} I_G(m_L) - \frac{N_{m_R}}{N_m} I_G(m_R)$ is the impurity reduction at node m , m_L and m_R are the left and right child nodes, and N is the total number of samples. The importance scores for all considered features were then normalized to sum to unity.

For a feature x_i and model $f(\mathbf{x})$, the partial dependence function is [19]

$$f_{x_i}(z) = \mathbb{E}_{\mathbf{x}_{-i}}[f(x_i = z, \mathbf{x}_{-i})] = \frac{1}{n} \sum_{j=1}^n f(z, \mathbf{x}_{j,-i}), \quad (8)$$

where $\mathbf{x}_{-i}$ denotes all features except x_i .

In this study, a discrete approximation of the partial dependence function was employed using quantile binning Q_k into ten bins. For each quantile, the average target value and feature value were calculated:

$$\bar{y}_k = \frac{1}{|Q_k|} \sum_{i \in Q_k} y_i, \quad (9)$$

$$\bar{x}_{i,k} = \frac{1}{|Q_k|} \sum_{i \in Q_k} x_i. \quad (10)$$

Sensitivity was then assessed as

$$S = \frac{\Delta \bar{y}}{\Delta \bar{x}_i} = \frac{\max_k(\bar{y}_k) - \min_k(\bar{y}_k)}{\max_k(\bar{x}_{i,k}) - \min_k(\bar{x}_{i,k})}, \quad (11)$$

which approximates the average marginal effect $\partial \mathbb{E}[y|x_i]/\partial x_i$.

D. Particle count verification

Correlations for pressure drop require the porosity value of the packed bed. However, loose random packings exhibit inherent porosity variations, making precise determination challenging

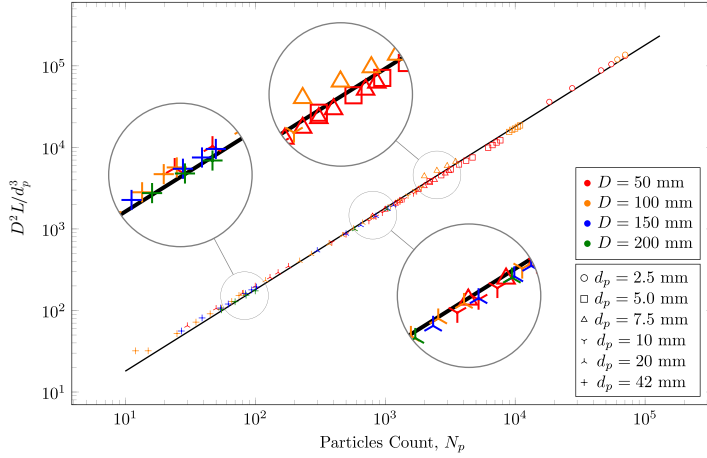


FIG. 3. Coherence verification of the particle enumeration method for all experimental samples. The total particle count N_p is plotted as a function of the dimensionless geometric group D^2L/d_p^3 , which represents the packed-bed volume normalized by the particle volume.

even for uniform spherical particles without knowing the exact particle count. Therefore, the number of particles was estimated for each packing configuration considered.

The enumeration of particles was performed using a dual approach. Larger particles were counted manually. For smaller particles, where counts exceeded 100, a gravimetric method was employed. This involved determining the average mass of a single particle by weighing a known subsample and then calculating the total count from the mass of the entire sample. All mass measurements were performed using a laboratory balance (HCS3003HD, resolution ± 0.05 g).

To ensure the accuracy of the calculated particle counts across all experimental configurations, a coherence test was implemented based on the geometric relationship between particle count and packed-bed volume. For uniformly sized particles in a cylindrical column, the total particle count N_p is theoretically proportional to the dimensionless group $\frac{D^2L}{d_p^3}$, where D is the column inner diameter (in millimeters), L is the packed-bed length (in millimeters), and d_p is the particle diameter (in millimeters).

This dimensionless group represents the packed-bed volume (proportional to D^2L) normalized by the particle volume (proportional to d_p^3). In the ideal case of perfect packing with uniform spheres, the relationship becomes

$$N_p = \varepsilon \frac{\pi}{6} \frac{D^2L}{d_p^3}, \quad (12)$$

where ε is the bed porosity.

Figure 3 represents the total particle count N_p versus the dimensionless geometric group D^2L/d_p^3 (packed-bed volume normalized by particle volume) on logarithmic axes. Data points represent individual experimental configurations with different symbols for particle diameters $d_p = 2.5, 5.0, 7.5, 10, 20, 42.6$ mm and colors for column diameters $D = 50, 100, 150, 200$ mm. The close alignment along the ideal proportionality line validates the counting methodology: Larger particles ($d_p = 42.6$ mm) were counted manually, while smaller particles ($d_p \leq 20$ mm) were enumerated gravimetrically using mass measurements with ± 0.05 g resolution. Deviations from the linear trend, particularly for large $d_p = 42.6$ mm particles at low N_p , would indicate potential counting errors requiring recalculation.

The effect of key packed-bed design parameters on porosity was analyzed to validate the experimental data. A weak negative linear correlation was observed between the dimensionless column

length and porosity, attributed to the fixed contribution of the lower-density region adjacent to the perforated plates: $\varepsilon \propto -L/D$. The relative particle size governs the thickness of the low-density layer near the wall and therefore exhibits a stronger, approximately quadratic relationship with porosity. For the present dataset, this correlation is given by

$$\varepsilon = 0.628 + 0.0025(D/d_p)^2 - 0.0456(D/d_p) - 0.0046(L/D). \quad (13)$$

E. Handling of outliers

The data cleaning involved the detection and removal of outliers. This was performed by fitting an ordinary least-squares regression to the log-transformed data: superficial velocity U_s^2 and the pressure drop ΔP . The method was selected for its consistency with the power-law assumption and computational efficiency. The detection itself relied on standardized residuals (z -scores z_i) to enhance reliability and ensure comparability, rather than on raw residuals

$$z_i = \frac{x_i - \bar{x}}{\sigma_x}, \quad (14)$$

where x_i is a feature value, $\bar{x}$ is the mean of the feature values, and σ_x is the standard deviation. In the context of residual analysis for outlier detection, the relevant feature is the regression residual.

Outliers were processed manually in a two-stage procedure. The first stage involved correcting unambiguous data acquisition errors, such as misplaced decimal points or incorrect trailing zeros. In the second stage, adjustments were made to individual outliers that exhibited a clear misalignment with the overall data trend, provided that they were part of a redundant parameter pair.

This second-stage correction was applied only to singular outliers that were separated by at least four consecutive, well-behaved data points. The requirement for parameter redundancy is illustrated by the pressure drop and velocity measurements, which were recorded as both minimum and maximum values from the manometer or anemometer. These two values form a redundant pair, as they are expected to follow the same trend. Consequently, if one value in the pair was obviously misaligned with this trend and the other was consistent with the surrounding data, the former could be corrected with confidence.

A tiered threshold approach was implemented for outlier management using multiple sigma levels 2σ , 2.5σ , and 3σ to provide a graduated classification. This scheme categorized 95.4% of the data as inliers. A three-tiered protocol was then applied to the remaining outliers.

First-tier outliers, defined as those within 2–2.5 standard deviations ($2 \leq \sigma_{\text{res}} \leq 2.5$), were processed with the utmost conservatism. The first stage of processing (correcting unambiguous data acquisition errors) was rarely fruitful for this group. The second stage (trend-based adjustment) was applied only if an outlier was strictly isolated (separated by at least four consecutive, well-behaved data points) and belonged to a redundant parameter pair. Consequently, most first-tier outliers were retained in their original state to preserve data integrity.

Second-tier outliers (residuals between 2.5σ and 3σ) underwent the most extensive processing. They were occasionally corrected in the first stage due to clear data acquisition errors. For the second stage, the criteria of redundancy and isolation were still required; however, the requirement for isolation was applied more leniently, allowing physical coherence and the overall data trend to dominate strict statistical rules. Following the correction, most of these outliers were reclassified into the first-tier group and retained in the dataset.

Third-tier outliers (residuals exceeding 3σ) were almost always the result of data acquisition errors and were typically corrected to become inliers during the first processing stage. Any remaining third-tier outliers were subsequently processed according to the criteria of their new lower-tier classification following the initial correction.

Overall, outliers constituted 4.6% of the total data. Of these, 54% were retained without modification and 12% were discarded following the processing protocol.

Graphical analysis of the residuals (see the example in Fig. 4) was conducted for each series to evaluate deviation patterns and their physical plausibility. The majority of deviations were

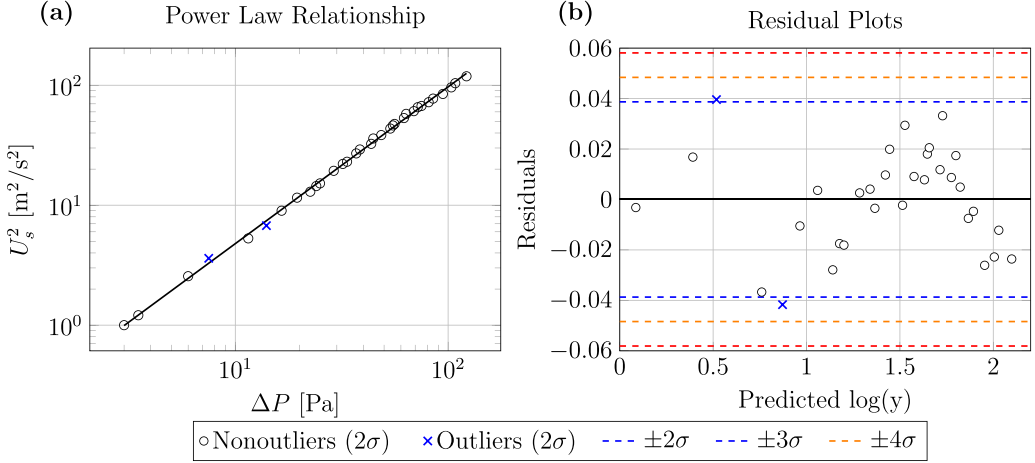


FIG. 4. Outlier detection and processing methodology demonstrated for pressure drop measurements with 5-mm particles in a 100-mm-diam pipe: (a) a log-log plot of pressure drop ΔP versus superficial velocity U_s showing the power-law relationship and ordinary least-squares regression line and (b) standardized residuals (z -scores) plotted against U_s , illustrating the three-tiered classification system with thresholds at $\pm 2\sigma$, $\pm 2.5\sigma$, and $\pm 3\sigma$ (dashed lines).

characterized by an increased scatter at the lower end of the superficial velocity range for each data series. We attribute this to the fixed absolute accuracy of the measurement instrumentation, whereby rounding errors become more significant relative to the smaller measured values of pressure and velocity.

In Fig. 4 presents outlier detection and processing methodology for pressure drop measurements with 5-mm particles in a 100-mm-diam pipe. Figure 4(a) shows a log-log plot of pressure drop ΔP versus superficial velocity U_s showing the power-law relationship and ordinary least-squares regression line. Data points are color coded by their outlier classification tier. Figure 4(b) depicts standardized residuals (z -scores) plotted against U_s , illustrating the three-tiered classification system with thresholds at $\pm 2\sigma$, $\pm 2.5\sigma$, and $\pm 3\sigma$ (dashed lines). Different symbols represent the processing outcomes: retained without modification (circle), corrected (square), or discarded (triangle). The methodology employed conservative processing: First-tier outliers ($2\sigma \leq |z| < 2.5\sigma$) were typically retained; second-tier outliers ($2.5\sigma \leq |z| < 3\sigma$) underwent the most extensive processing; third-tier outliers ($|z| \geq 3\sigma$) were usually corrected as data acquisition errors. Note the increased residual scatter at lower velocities due to fixed instrument accuracy.

F. Residual heat maps

The described residuals were calculated from a comparison with regressed trends such that the experimental data were used for self-verification. Similarly, the residuals were calculated from comparisons with well-known semiempirical correlations: Carman-Kozeny [1], Ergun [2], and Reger [20]. The normalized pressure drop in a packed bed is known to be proportional to a friction factor F_s , which can be estimated from the modified Reynolds number $\text{Re}_p = f(\varepsilon)$ using a set of model constants K_1 , K_2 , and n :

$$\frac{\Delta P}{L} = F_s \frac{\rho U_s^2}{d_p} \frac{1 - \varepsilon}{\varepsilon^3}, \quad (15)$$

$$F_s = \frac{K_1}{\text{Re}_p} + \frac{K_2}{\text{Re}_p^n}, \quad (16)$$

TABLE II. Model constants for estimating the friction factor F_s .

Equation	K_1	K_2	n
Carman [1]	180	2.87	0.1
Ergun [2]	150	1.75	0
Reger <i>et al.</i> [20]	160	$3f(\varepsilon)$	0.1

$$\text{Re}_p = \frac{\rho U_s d_p}{\mu(1 - \varepsilon)}, \quad (17)$$

$$\varepsilon = \frac{V_{\text{voids}}}{V} = 1 - \frac{V_p}{V} = 1 - \frac{2d_p^3 N}{3D^2 L}. \quad (18)$$

Model constants vary considerably across published studies (see Table II), but their net effect on the results is minimal. For example, the model of Reger *et al.*, which employs a high-order polynomial $f(\varepsilon) = 253.9\varepsilon^4 - 499.3\varepsilon^3 + 364.7\varepsilon^2 - 115.6\varepsilon + 14.21$ to compute K_2 as a function of porosity [20] and is specifically fine-tuned for low D/d_p values [5] (similar to the Reichelt equation [21]), yields results nearly identical to those of other models when applied to our experimental dataset. Based on its superior combination of simplicity and accuracy, the Carman model was selected for all subsequent analysis.

The regression residuals were visualized using heat maps to localize the segments of high prediction error. The heat maps illustrate the spatial distribution of the mean absolute relative error (MARE) across the domain of normalized geometric parameters. The bed diameter was normalized by the particle diameter D/d_p and the bed length was normalized by the bed diameter L/D . The MARE for each data point was calculated as

$$\mathcal{M} = \frac{1}{N} \sum_{i=1}^N \frac{|\Delta P_{\text{expt } i} - \Delta P_{\text{model } i}|}{\Delta P_{\text{expt } i}}. \quad (19)$$

III. RESULTS AND DISCUSSION

A. Raw dataset

The raw dataset, comprising over 3000 measured points, was analyzed to evaluate the MARE against the models listed in Table II. Additionally, a sensitivity analysis was performed to quantify the influence of the following dimensionless parameters on the MARE: the superficial velocity U_s , the aspect ratio L/D , and the diameter ratio D/d_p . The residual heat maps for the raw dataset are presented in Fig. 5.

Figure 5 reveals significant deviations between experimental results and model predictions. The strongest deviation occurs for the monolayer configuration, where 42.6-mm spheres are placed in a 50-mm tube. In this case, the packed bed exhibits high porosity due to wall effects. This configuration represents a limiting case rather than a typical packed bed, as the diameter ratio $D/d_p < 2$ falls outside the conventional range for which most pressure drop correlations are validated.

Linear correlation analysis yields weak relationships between the residuals and the dimensionless parameters, with Pearson correlation coefficients of $\rho_{D/d_p} = -0.2$, $\rho_{L/D} = 0.08$, and $\rho_{U_s} = 0.18$. However, nonlinear sensitivity analysis, based on feature importance metrics, reveals a more balanced influence of the three parameters: $\phi_{D/d_p} = 42.3\%$, $\phi_{L/D} = 37.7\%$, and $\phi_{U_s} = 20.0\%$. These results indicate that for the raw dataset, spanning a wide range of operating and design conditions, no single parameter dominates the prediction accuracy of pressure drop models. Instead, the prediction error appears to be collectively influenced by all three variables, underscoring the complexity of accurately modeling packed beds across diverse geometrical and flow regimes.

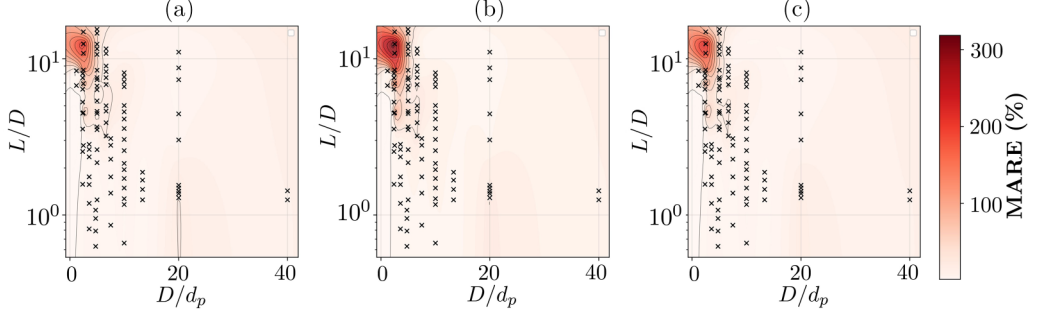


FIG. 5. Residual heat maps for the raw dataset, showing deviations between experimental measurements and model predictions from (a) Carman [1], (b) Ergun [2], and (c) Reger *et al.* [20]. The color scale represents the magnitude of the relative error.

B. Data cleaning

1. First iteration

Analysis of the raw dataset indicated that significant residuals occur at low D/d_p ratios, particularly for the 50-mm tube filled with 42.6-mm spheres. To sanitize the data, an inner cut was applied, requiring each series to satisfy at least one of two criteria: $D/d_p \geq 4$. Series failing both conditions were excluded. The remaining data formed the core dataset, comprising 63% of the original measurements.

The residual heat maps in Fig. 5 exhibit similar behavior across all models considered. For subsequent analysis, only the Ergun equation was used, as it is widely adopted in both literature and industrial applications. Figure 6 shows the residual heat map for the cleaned dataset after the first iteration.

After cleaning, the superficial velocity exhibited the strongest (though moderate) linear correlation with the residual, but with an inverse sign: $\rho_{U_s} = -0.39$. The geometric parameters showed weaker positive correlations: $\rho_{D/d_p} = 0.29$ and $\rho_{L/d_p} = 0.28$. Feature importance analysis revealed that the column length ratio L/d_p dominated the nonlinear influence, accounting for $\phi_{L/d_p} = 58\%$ of the importance, while the contributions of velocity and column diameter were more subdued: $\phi_{U_s} = 23.8\%$ and $\phi_{D/d_p} = 18.2\%$, respectively.

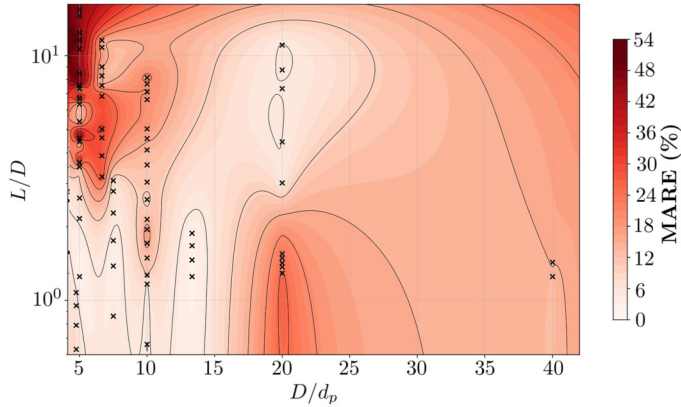


FIG. 6. Residual heat map after the first cleaning iteration. The color scale indicates the magnitude of relative error between experimental measurements and predictions from the Ergun equation.

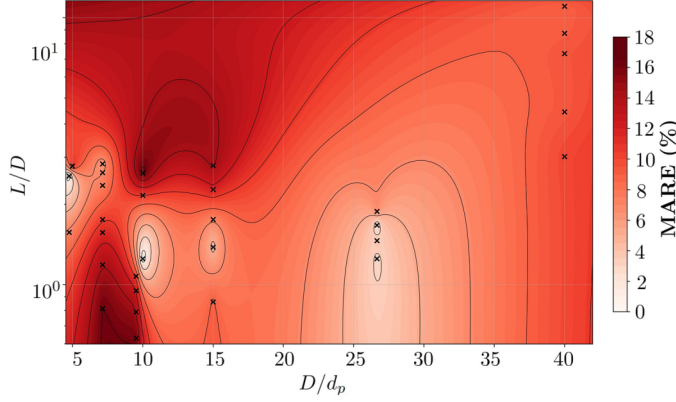


FIG. 7. Residual heat map after the second cleaning iteration. The color scale indicates the relative error between experimental measurements and predictions from the Ergun equation.

The increased importance of the L/D ratio can be explained by wall effects. For the cleaned dataset, significant wall influence on pressure drop persists at low D/d_p ratios. While increasing the L/D ratio reduces measurement error in determining specific pressure drop (Pa/m), low D/d_p ratios amplify wall effects. This observation aligns with findings reported by Dixon [22].

Additionally, increased residuals are observed for low L/D ratios even at high D/d_p ratios. In these configurations, the first layer of the packed bed contributes disproportionately to the overall pressure drop compared to subsequent layers. Consequently, experimental measurements at low L/D ratios yield higher pressure drops than those predicted by Ergun equation.

2. Second and third iterations

Although the cleaned data showed significant improvements in median residual values (see Fig. 6), nonuniformity in the residual distribution persisted. A second sensitivity analysis was therefore performed to further refine the dataset, isolating flow regions where the Ergun approximation remains valid under the experimental conditions.

The resulting dataset after the second cleaning iteration (Fig. 7) comprises 30 experimental series, representing 32% of the initial data. For this subset, the discrepancy between experimental measurements and predictions from the Ergun (Carman) equation ranges from 4% to 18%, with an average residual of 12%. The parameter space covered spans D/d_p from 2.5 to 20 and L/D from 3 to 12. No discernible pattern or trend was observed in the residual distribution across this range.

In terms of linear correlation, superficial velocity remains the strongest and most consistent parameter, showing a moderate inverse relationship with the residual: $\rho_{U_s} = -0.40$. The geometric parameters now exhibit only negligible linear correlations, $\rho_{L/d_p} = 0.04$ and $\rho_{D/d_p} = -0.13$, indicating that their influence is predominantly nonlinear. This observation is confirmed by feature importance analysis, which identifies superficial velocity as the dominant nonlinear driver, accounting for $\phi_{U_s} = 50.1\%$ of the model's predictive power. The length-to-diameter ratio L/D remains an important secondary factor ($\phi_{L/D} = 35.1\%$), while the diameter ratio D/d_p plays a smaller role ($\phi_{D/d_p} = 14.8\%$).

The shift in feature importance, where velocity becomes more influential while L/D and D/d_p become less so, can be explained by two factors. First, the Ergun equation assumes constant coefficients A and B for the kinetic and viscous terms across all flow regimes. However, it is well established that the viscous term is not constant and depends on the flow regime, as first observed by Nikuradse [23]. Second, despite the reduction in importance, the L/D ratio still notably affects the deviation between experimental and predicted results. To ensure thermal uniformity

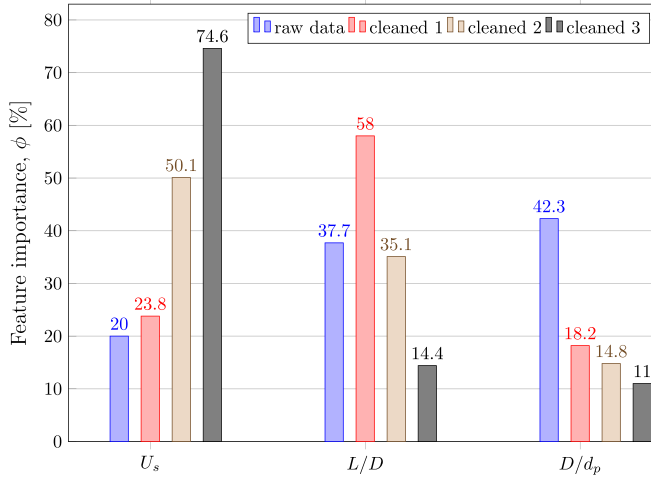


FIG. 8. Feature importance for the raw and cleaned dataset.

and minimize wall effects within the packed bed, the column diameter must be sufficiently large, requiring $L/D > 5$ [3,24].

To further reduce wall effects, a third cleaning iteration was performed to select data with moderately high L/D and D/d_p ratios. This subset primarily corresponds to packed beds of 100 and 150 mm diameter filled with 7.5-, 10-, and 20-mm spheres.

Results after the third iteration remained consistent with previous findings. Superficial velocity maintained the strongest linear relationship with the output ($\rho_{U_s} = -0.37$), while geometric parameters exhibited weaker linear correlations: $\rho_{L/d_p} = 0.08$ and $\rho_{D/d_p} = 0.18$. In the nonlinear model, superficial velocity dominated with a feature importance of $\phi_{U_s} = 74.6\%$, while the relative column length ($\phi_{L/D} = 14.4\%$) and diameter ratio ($\phi_{D/d_p} = 11.0\%$) showed nearly equal importance.

This evolution strongly suggests that the final filtering step successfully mitigated noise and confounding effects, yielding a more reliable and interpretable model where superficial velocity, the dominant physical driver, is consistently identified across all metrics.

C. Normalized importance

Figure 8 presents the normalized Gini importance ϕ for the three dimensionless parameters, i.e., the superficial velocity U_s , length-to-diameter ratio L/D , and diameter ratio D/d_p , across the raw dataset and the three successive cleaning iterations. The evolution of feature importance clearly illustrates how systematic data refinement isolates the dominant physical driver of pressure drop prediction error.

For the raw dataset (first bar group), the three parameters exhibit relatively balanced influence, with D/d_p showing the highest importance ($\phi_{D/d_p} = 42.3\%$), followed by L/D ($\phi_{L/D} = 37.7\%$) and U_s ($\phi_{U_s} = 20.0\%$). This distribution reflects the complex multifaceted nature of errors when including configurations with strong wall effects (particularly $D/d_p < 2$) and entry or exit effects.

After the first cleaning iteration (second bar group), which removed configurations with $D/d_p < 4$, a significant shift occurs: L/D becomes the dominant factor ($\phi_{L/D} = 58.0\%$), while D/d_p importance drops substantially ($\phi_{D/d_p} = 18.2\%$). The increased relative importance of L/D suggests that with extreme wall effects removed, length-dependent phenomena, such as the disproportionately high pressure drop contribution from the first particle layer at low L/D ratios, become more prominent in explaining residual errors.

The second cleaning iteration (third bar group), which further refined the dataset to regions where the Ergun approximation is more likely to be valid, reveals another transition: U_s emerges as the

dominant parameter ($\phi_{U_s} = 50.1\%$), while L/D importance decreases ($\phi_{L/D} = 35.1\%$) and D/d_p becomes the least influential ($\phi_{D/d_p} = 14.8\%$). This shift indicates that as geometric irregularities are progressively filtered out, flow-dependent factors, particularly the inadequacy of the Ergun equation's constant coefficients across different flow regimes, become the primary source of prediction error.

The third and final cleaning iteration (fourth bar group), which selected configurations with moderate to high L/D and D/d_p ratios to minimize wall effects, shows U_s achieving overwhelming dominance ($\phi_{U_s} = 74.6\%$), with both geometric parameters reduced to nearly equal minor roles ($\phi_{L/D} = 14.4\%$ and $\phi_{D/d_p} = 11.0\%$). This progression demonstrates that after systematically removing data points affected by wall effects, entry and exit phenomena, and extreme geometries, the superficial velocity, the key parameter in the Ergun equation's kinetic and viscous terms, emerges unequivocally as the most important factor governing prediction accuracy.

The monotonic increase in U_s importance across iterations ($20.0\% \rightarrow 23.8\% \rightarrow 50.1\% \rightarrow 74.6\%$) and corresponding decrease in geometric parameter importance confirm that the cleaning protocol successfully isolated the core physical limitation of the Ergun model: its inability to accurately capture flow-regime-dependent behavior across the full range of experimental conditions. This finding has important implications for model improvement, suggesting that efforts should focus on developing velocity-dependent correction factors or regime-aware coefficients rather than refining geometric correction terms.

IV. CONCLUSION

A residual-driven sensitivity analysis framework has been developed and applied to systematically evaluate the influence of key parameters, i.e., the superficial velocity U_s , normalized bed length L/D , and normalized column diameter D/d_p , on the predictive accuracy of classical pressure drop models for randomly packed beds. Using an extensive experimental dataset collected in our laboratory and integrating robust outlier detection, multitiered data cleaning, and complementary statistical methods (Pearson correlation, random forest feature importance, and partial dependence analysis), this study provides several key insights.

First, the analysis of the raw dataset revealed that prediction errors are collectively influenced by all three parameters, with no single variable dominating. This underscores the complexity of modeling packed beds across a wide range of geometric and flow conditions, particularly when configurations with strong wall effects ($D/d_p < 2$) are included. The feature importance distribution for the raw data ($\phi_{D/d_p} = 42.3\%$, $\phi_{L/D} = 37.7\%$, and $\phi_{U_s} = 20.0\%$) reflects this multifaceted nature of error sources.

Second, systematic data refinement through three iterative cleaning steps successfully isolated the core physical driver of model discrepancy. As geometric irregularities and extreme configurations were progressively filtered out—first by removing data with $D/d_p < 4$, then by focusing on regimes where the Ergun approximation is valid, and finally by selecting configurations with moderate to high L/D and D/d_p ratios—the importance of superficial velocity increased monotonically from 20.0% to 74.6%, while geometric factors became correspondingly less influential. This clear progression demonstrates that the proposed cleaning protocol effectively mitigates confounding effects, yielding a more interpretable dataset.

Third, the emergence of superficial velocity as the overwhelmingly dominant parameter in the final cleaned dataset ($\phi_{U_s} = 74.6\%$) points to a fundamental limitation of widely used correlations like the Ergun equation: their inability to accurately capture flow-regime-dependent behavior. The assumption of constant coefficients for viscous and inertial terms across all Reynolds numbers appears inadequate, particularly as geometric perturbations are minimized. This finding suggests that efforts to improve pressure drop predictions should focus on developing velocity-dependent correction factors or regime-aware coefficients, rather than further refining geometric correction terms.

The methodology presented, that is, combining rigorous outlier handling, parameter-space mapping via residual heat maps, and multifaceted sensitivity assessment, provides a robust framework for analyzing experimental fluid dynamics data. Beyond the specific findings for packed beds, this approach offers practical guidance for data-quality assurance and model selection in applications where predictive accuracy depends on multiple interacting parameters.

Future work should explore the implementation of velocity-dependent correction functions in existing semiempirical models and validate the proposed sensitivity analysis framework on independent datasets from different experimental setups. Extending this approach to other complex multiphase flows could further demonstrate its utility in identifying dominant error sources and guiding model improvement.

DATA AVAILABILITY

The data that support the findings of this article are openly available [25]; embargo periods may apply.

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